Aptitude & Logical Reasoning

Ultimate Shortcut Methods & Formula Guide

Set 1 — Quantitative Aptitude & Reasoning

Q1. Smallest number added to 1,789 to make it divisible by 11.

Shortcut: Divide 1789 by 11 → Remainder is 7. Required number = 11 - 7 = **4**.

Q2. Smallest 4-digit number > 2,000 ending in 5 and divisible by 9.

Shortcut: Form format 2XYZ5 → Smallest starts 20_5. Sum of digits for 2+0+X+5 must be a multiple of 9. 7 + X = 9 → X = 2. Number is **2,025**.

Q3. 15 students' average increases by 3 when 37 is replaced. New score?

Shortcut: New = Old + (Total Students × Increase) = 37 + (15 × 3) = 37 + 45 = **82**.

Q4. Pipe A fills in 6h, B empties in 8h. Alternating 1h each (starting with A). Time to fill?

Shortcut: Net 2-hour work = 1/6 - 1/8 = 1/24. To avoid overflow, target work before the last stroke: Total capacity = 24 units. A fills 4, B drains 3. Target = 24 - 4 = 20 units. 20 units take 20 × 2 = 40 hours. Remaining 4 units filled by A in 1 hour. Total time = 40 + 1 = **41 hours**.

Q5. Sum becomes 1.5 times in 8 years at simple interest. Rate?

Shortcut: Interest = 0.5 × Principal. Rate = (Interest × 100) / (P × T) = (0.5 × 100) / 8 = **6.25%**.

Q6. 3 machines produce 240 components in 4 hours. 5 machines in 6 hours?

Shortcut: Formula: (M1 × T1) / W1 = (M2 × T2) / W2 → (3 × 4) / 240 = (5 × 6) / W2 → 12 / 240 = 30 / W2 → W2 = 30 × 20 = **600 components**.

Q7. Fair die rolled twice. Probability of at least one six?

Shortcut: Total outcomes = 36. Outcomes with NO six = 5 × 5 = 25. P(at least one six) = 1 - 25/36 = **11/36**.

Q8. Letter Series: B, E, J, Q, ?

Shortcut: Positions: B(2), E(5), J(10), Q(17). Gaps are consecutive odd numbers (+3, +5, +7). Next gap is +9. 17 + 9 = 26 → **Z**.

Q9. Priya ranks 12th from top, 19th from bottom. Total students?

Shortcut: Total = Top + Bottom - 1 = 12 + 19 - 1 = **30 students**.

Q10. Code SMART with +3 shift.

Shortcut: S+3=V, M+3=P, A+3=D, R+3=U, T+3=W → **VPDUW**.

Data Interpretation: Selection Analysis

College	Applicants	Selected	Percentage

A	120	72	60.0%
B	150	105	70.0%
C	180	117	65.0%
Overall	450	294	65.33%

Q12. Highest selection rate: College B (70%).

Q13. Overall selection rate: (294 / 450) × 100 = **65.33%**.

Set 2 — Advanced Methods

Q1. If 40% of A equals two-thirds of B, ratio A:B?

Shortcut: 0.4A = (2/3)B → (2/5)A = (2/3)B → A/B = (2/3) × (5/2) = 5/3 → **A:B = 5:3**.

Q2. AP: 7th term = 31, 13th term = 55. Find first term (a) and diff (d).

Shortcut: Shortcut for d: (T13 - T7) / (13 - 6) → (55 - 31) / 6 = 24 / 6 = **4**. First term a = T7 - 6d = 31 - 6(4) = **7**.

Q3. Average of 5 numbers is 24. Four numbers are 18, 20, 25, 27. Find 5th.

Shortcut: Deviation method (Target 24): Gaps are -6, -4, +1, +3. Net deviation = -6. Fifth number must balance this: 24 + 6 = **30**.

Q4. Clock gains 5 mins/hour. Set at 8:00 AM. Time at 2:00 PM?

Shortcut: Duration = 6 hours. Total gain = 6 × 5 = 30 mins. Time shown = **2:30 PM**.

Q5. CI on ■10,000 for 6 months at 20% p.a. compounded quarterly.

Shortcut: Quarterly rate = 20% / 4 = 5%. Period = 2 quarters. Effective standard rate = 5 + 5 + (5×5)/100 = 10.25%. CI = 10.25% of 10,000 = **■1,025**.

Q6. Race: A beats B by 20m in 100m. B beats C by 25m in 100m. A beats C by?

Shortcut: When A=100, B=80. When B=100, C=75. Ratio C/A = (C/B) × (B/A) = (75/100) × (80/100) = 60/100. Thus, when A=100, C=60. A beats C by 100 - 60 = **40m**.

Q7. Two cards drawn simultaneously from 52 cards. P(both are aces)?

Shortcut: P = (4/52) × (3/51) = (1/13) × (1/17) = **1/221**.

Q8. Number of factors of 720.

Shortcut: Prime factorise: 720 = 2^4 × 3^2 × 5^1. Total factors = (4+1)(2+1)(1+1) = 5 × 3 × 2 = **30 factors**.

Q9. Ordering rule: P before Q, R before P, Q before S.

Shortcut: Chain sequence: R → P → Q → S. Valid order: **RPQS**.

Q10. Sales: Jan 2.4L, Apr 4.2L. % Increase & Average?

Shortcut: Increase = (4.2 - 2.4)/2.4 × 100 = (1.8/2.4) × 100 = **75%**. Average = (2.4 + 3.0 + 3.6 + 4.2) / 4 = 13.2 / 4 = **■3.3 lakh**.

Set 3 — Fast Calculations & Logic

Q1. A, B, C share ■18,000. A does 1/2, B and C share remaining 1/2 equally.

Shortcut: Shares: A = 1/2 = 50%, B = 25%, C = 25%. A gets **■9,000**, B gets **■4,500**, C gets **■4,500**.

Q2. Smallest number leaving remainder 2 mod 5, 3 mod 7, 4 mod 9.

Shortcut: Difference between divisor and remainder is constant: 5-2=3, 7-3=4 (not constant). Let's check values: N = 9k + 4. For k=17, 9(17)+4 = 157. Verify: 157 % 5 = 2; 157 % 7 = 3. **N = 157**.

Q3. 4 White, 6 Black balls. Two drawn with replacement. P(White then Black)?

Shortcut: P = P(W) × P(B) = (4/10) × (6/10) = 24/100 = **6/25 (or 0.24)**.

Q4. Arrangements of LEVEL beginning and ending with L.

Shortcut: Fix L at start and end: L [E, V, E] L. Remaining letters rearrange as 3! / 2! (since E repeats) = **3 ways**.

Q5. A invests 40k for 12 months, B invests 60k for 8 months. Profit share?

Shortcut: Ratio of investment-months: A : B = (40,000 × 12) : (60,000 × 8) = 480k : 480k = 1:1. Split total ■26,000 equally: **■13,000 each**.

Q6. 12 pumps empty in 15h. How long for 20 pumps?

Shortcut: M1 × D1 = M2 × D2 → 12 × 15 = 20 × T2 → 180 = 20 × T2 → T2 = **9 hours**.

Q7. Evaluate (999^2 - 998 × 1000).

Shortcut: Algebraic Identity: x^2 - (x-1)(x+1) = x^2 - (x^2 - 1) = **1**.

Q8. Series: 1, 2, 6, 24, 120, ?

Shortcut: Factorials: 1!, 2!, 3!, 4!, 5!. Next is 6! = 120 × 6 = **720**.

Q9. Ordering: A before C, D before B, E before A.

Shortcut: Merge rules: E → A → C and D → B. One valid combined string: **EDACB**.

Q10. Data Sufficiency for integer n. I: Divisible by 6. II: < 10 and divisible by 4.

Shortcut: From II, positive single digits divisible by 4 are 4 and 8 (not unique). I gives infinitely many multiples. Combine I & II: no single positive integer satisfies both < 10, div by 4 and div by 6. Statements are **Insufficient**.

Q11. Find median score: 72, 84, 66, 90, 78.

Shortcut: Arrange in ascending order: 66, 72, **78**, 84, 90. Middle term is **78**.

Q12. Score 90 is increased by 10% then reduced by 10%. New value?

Shortcut: Net change formula = x - y - xy/100 → 10 - 10 - 1 = -1% (net loss). 90 × 0.99 = **89.1**.

Set 4 — Equations & Speed Matrix

Q1. Marked up 40%, success discounts 10% and 15%. Net profit %?

Shortcut: Base 100 → Mark up = 140. Discount 10% = 140 - 14 = 126. Discount 15% = 126 - 18.9 = 107.1. Net profit = **7.1%**.

Q2. 180m train crosses pole in 12s. Time to cross a 420m platform?

Shortcut: Speed = 180/12 = 15 m/s. Total distance to cross platform = 180 + 420 = 600m. Time = 600 / 15 = **40 seconds**.

Q3. A in 12 days, B in 18 days. Together 3 days, then A leaves. Extra days for B?

Shortcut: LCM of 12, 18 = 36 units. Efficiency A = 3, B = 2. Together 3 days = (3+2)×3 = 15 units. Remainder = 36 - 15 = 21 units. B takes 21 / 2 = **10.5 additional days**.

Q4. Smallest positive integer leaving rem 5 mod 8 and rem 9 mod 12.

Shortcut: Divisor minus remainder: 8-5=3; 12-9=3. Since difference (3) is constant: LCM(8, 12) - 3 = 24 - 3 = **21**.

Q5. Difference between CI and SI for 2 years at 10% is ■50. Principal?

Shortcut: Shortcut: Diff = P(R/100)^2 → 50 = P(10/100)^2 → 50 = P(1/100) → P = **■5,000**.

Q6. Two dice rolled. Probability of sum ≥ 9?

Shortcut: Favorable sums (9,10,11,12) pairings count = 4 (for 9) + 3 (for 10) + 2 (for 11) + 1 (for 12) = 10. P = 10/36 = **5/18**.

Q7. 3-digit numbers divisible by 5 using digits 1,2,3,4,5 without repetition.

Shortcut: Last digit must be 5 (1 choice). Remaining 2 slots filled by remaining 4 digits: 4 × 3 = **12 numbers**.

Q8. Series: 3, 7, 15, 31, 63, ?

Shortcut: Pattern: (Previous × 2) + 1 → (63 × 2) + 1 = 126 + 1 = **127**.

Q11. 8 people handshakes total count.

Shortcut: Formula: n(n-1)/2 → 8 × 7 / 2 = **28 handshakes**.

Q12. Angle between hands at 4:20.

Shortcut: Formula: |30H - 5.5M| → |30(4) - 5.5(20)| = |120 - 110| = **10°**.

Q13. A fills 10h, B fills 15h, Leak drains 30h. Combined time?

Shortcut: LCM of 10, 15, 30 = 30 units. Rates: A=+3, B=+2, Leak=-1. Net rate = 3 + 2 - 1 = +4 units/hour. Time = 30 / 4 = **7.5 hours**.

Q14. Sales: Q1 4.8L, Q4 7.8L. % Increase?

Shortcut: Increase = (7.8 - 4.8) / 4.8 × 100 = 3 / 4.8 × 100 = **62.5%**.

Q15. Target average 80 for 6 tests. First 5 scores: 72, 84, 68, 76, 90.

Shortcut: Total needed = 80 × 6 = 480. Sum of first 5 = 390. Sixth score = 480 - 390 = **90**.